

Toothbrush Holder

3D Printed Parts

1x Main body (PLA) This is the main housing for all the parts it also has two tabs that help lock the toothbrush in place.

2x Side doors (PLA) This shields the toothbrush from debris and stops UVC light from getting out.

2x Rods (PLA) Helps lock the toothbrush in and helps with the movement of the doors.

1x Band (TPU) Helps to grip the toothbrush and allows compatibility with most standard toothbrushes.

Electronics

Microcontroller

1xESP32(TBD) If I use this it will be for controlling the servos and running simple algorithms.

Effectors

1x UVC LED This is for sterilizing the toothbrush.

2x Servo For locking the side doors in the closed position.

1x Green LED For letting the user know the UVC led is on.

1x Red LED for letting the user know something is wrong its basically an error message

Sensors

3x Buttons one For sensing when the door is pressed on. And another for when a toothbrush is inserted and one to check if the side doors are closed

Power source

TBD (Probably just going to plug it in for now).

Goal solar panel

